AN ISOMORPHISM OF COMPLETE LOCAL RINGS IS APPROXIMATED BY A DIAGRAM OF ELEMENTARY ETALE NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11070-11095 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 11097-11141, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. G-rings, complete local rings, elementary etale neighbourhoods, and approximation of maps between completions; the construction is retained.

Source reference. Original lemma lemma-lifting-along-artinian-at-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3575–3600. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-map-approximation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11017–11039. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AGX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11141–11141. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11141–11141. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let X, Y be schemes locally of finite type over S. Let $x \in X$ and $y \in Y$ be points lying over the same point $s \in S$. Assume $\mathcal{O}_{S,s}$ is a G-ring. Assume we have an $\mathcal{O}_{S,s}$-algebra isomorphism*

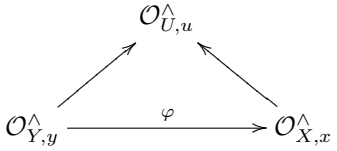
$$\varphi : \mathcal{O}^\wedge_{Y,y} \longrightarrow \mathcal{O}^\wedge_{X,x}$$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

between the complete local rings. Then for every $N \geq 1$ there exists morphisms

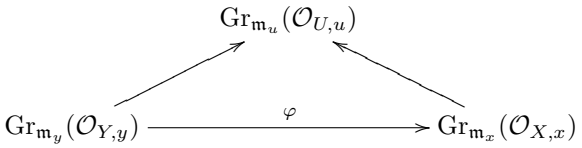
$$(X, x) \leftarrow (U, u) \rightarrow (Y, y)$$

of pointed schemes over S such that both arrows define elementary étale neighbourhoods and such that the diagram



commutes modulo $\mathfrak{m}_u^N$.

Proof. We may assume $N \geq 2$. Apply Lemma lemma-map-approximation to get $(U, u) \rightarrow (X, x)$ and $f : (U, u) \rightarrow (Y, y)$. We claim that f is étale at u which will finish the proof. In fact, we will show that the induced map $\mathcal{O}^\wedge_{Y,y} \rightarrow \mathcal{O}^\wedge_{U,u}$ is an isomorphism. Having proved this, Lemma lemma-lifting-along-artinian-at-point will show that f is smooth at u and of course f is unramified at u as well, so Morphisms, Lemma 02GK tells us f is étale at u. For a local ring $(R, \mathfrak{m})$ we set $\mathrm{Gr}_\mathfrak{m}(R) = \bigoplus_{n \geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1}$. To prove the claim we look at the induced diagram of graded rings



Since $N \geq 2$ this diagram is actually commutative as the displayed graded algebras are generated in degree 1! By assumption the lower arrow is an isomorphism. By More on Algebra, Lemma 0AGX (for example) the map $\mathcal{O}^\wedge_{X,x} \rightarrow \mathcal{O}^\wedge_{U,u}$ is an isomorphism and hence the north-west arrow in the diagram is an isomorphism. We conclude that f induces an isomorphism $\mathrm{Gr}_{\mathfrak{m}_x}(\mathcal{O}_{X,x}) \rightarrow \mathrm{Gr}_{\mathfrak{m}_y}(\mathcal{O}_{U,u})$. Using induction and the short exact sequences

$$0 \rightarrow \mathrm{Gr}^n_\mathfrak{m}(R) \rightarrow R/\mathfrak{m}^{n+1} \rightarrow R/\mathfrak{m}^n \rightarrow 0$$

for both local rings we conclude (from the snake lemma) that f induces isomorphisms $\mathcal{O}_{Y,y}/\mathfrak{m}_y^n \rightarrow \mathcal{O}_{U,u}/\mathfrak{m}_u^n$ for all n which is what we wanted to show. □